

Random Hyperbolic Surface Spectral Rigidity — cycles 4-6

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Random Hyperbolic Surface Spectral Rigidity — cycles 4-6

Abstract

Cycles 4-6 completed the proof-ledger phase for Kim-Tao’s random-cover spectral rigidity paper and began the first computational probe phase. Cycle 4 reconstructed the Theorem 2 eigenfunction delocalization argument from the twisted pre-trace formula through the fourth-moment statistic, diagonal subtraction, probability conversion, and final L^∞ bound. Cycle 5 consolidated the full M2-proof-ledger milestone by producing a unified Theorem 1 rigidity reconstruction, a cross-proof loss map, and a closure note accepted by audit. Cycle 6 opened M3-computational-probes with a reproducible random-permutation common-fixed-point benchmark.

The main decision made during these cycles was that M2-proof-ledger is now validated, narrowly for local proof reconstruction and quantitative dependency/loss accounting. The main new empirical result is a toy-level confirmation of the diagonal-subtraction mechanism: cyclic primitive-power word families have order-one common fixed-point counts, while rank-two pair families scale near $1/n$. The same experiment also found that naive multiword pointwise intersections are not a good model for the paper’s eight-word folded-graph mechanism, because several composite word families collapse to the same fixed-point constraint as the pair (a, b) .

Introduction

The campaign studies Elena Kim and Zhongkai Tao’s paper Eigenvalue rigidity of hyperbolic surfaces in the random cover model, using the local files 2603.01127.pdf and 2603.01127.txt. Earlier cycles established the paper map and reconstructed the trace-side proof of Theorem 1 through Proposition 3.1. Cycles 4-6 continued from that state.

The report uses the following terms.

- M2-proof-ledger is the milestone for reconstructing the paper’s main proof architecture, quantitative dependencies, and exponent losses.

- M3-computational-probes is the milestone for reproducible finite or toy computations that test mechanisms exposed by the proof ledger.
- V_n is the fourth power of the centered local spectral mass appearing in the Theorem 2 pre-trace argument.
- S is the primitive-power diagonal term subtracted from V_n before applying the rank-two common-fixed-point estimate.
- α_W is the exponent in the uniform Weyl law; α_R is the possibly weaker eigenvalue-rigidity exponent after Weyl-law inversion near the spectral edge $\lambda = 1/4$.

Approach

The reporting pass gathered the specified cycle sessions, local artifacts, ledger events, figures, generated datasets, validation outputs, and the current workspace manifest. The source set included all nine requested cycle sessions:

- Cycle 4: researcher 9b69e544-cee2-427e-88a7-a2b32be2d4df, worker 9a8d5232-7da5-416e-be1a-8515bcece60c, auditor dddcaf65-1172-4fa3-a859-9374ae0213b4.
- Cycle 5: researcher 0bf32ed4-2866-4671-af3c-2f1b6f30f8ba, worker b5a3c22f-3083-4803-b8c6-1f8e8dee3dc2, auditor 7dc50143-c79a-4b3c-8371-63f8e61aa904.
- Cycle 6: researcher 40632ca0-1b63-4d2b-9bc8-c8baab1f4ae4, worker fe7f867c-00e3-47b8-a4cd-172c205ba885, auditor 3229364f-0203-4038-8c82-32992a28250f.

No REFERENCES.md file was present in the workspace, so this report cannot continue a global numbered bibliography. Sources are therefore cited by local paper file, artifact path, and session ID.

Source Inventory and Timeline

Cycle 4: Theorem 2 Delocalization Reconstruction

The Cycle 4 researcher brief identified the remaining proof-ledger gap as the eigenfunction side of the paper. The worker was directed to reconstruct Theorem 2 from the twisted pre-trace formula, define V_n and S , explain Proposition 4.1 and Proposition 4.2, and build a dependency graph.

The worker produced six artifacts:

- docs/proof_ledger/delocalization_proof_reconstruction.md
- docs/proof_ledger/eigenfunction_fourth_moment_ledger.md
- docs/proof_ledger/pretrace_diagonal_term.md
- docs/proof_ledger/theorem2_dependency_graph.dot
- docs/proof_ledger/theorem2_dependency_graph.png
- docs/proof_ledger/theorem1_theorem2_loss_comparison.md

The dependency graph was rendered successfully.

The Cycle 4 audit validated the mathematics of the Theorem 2 slice. It also found one structured-state defect: the worker had briefly marked broad M2-proof-ledger as validated even though the plan still required final proof-ledger consolidation. The auditor fixed this by reopening M2-proof-ledger and marking it in-progress, preserving the Theorem 2 slice validation without closing the broad milestone.

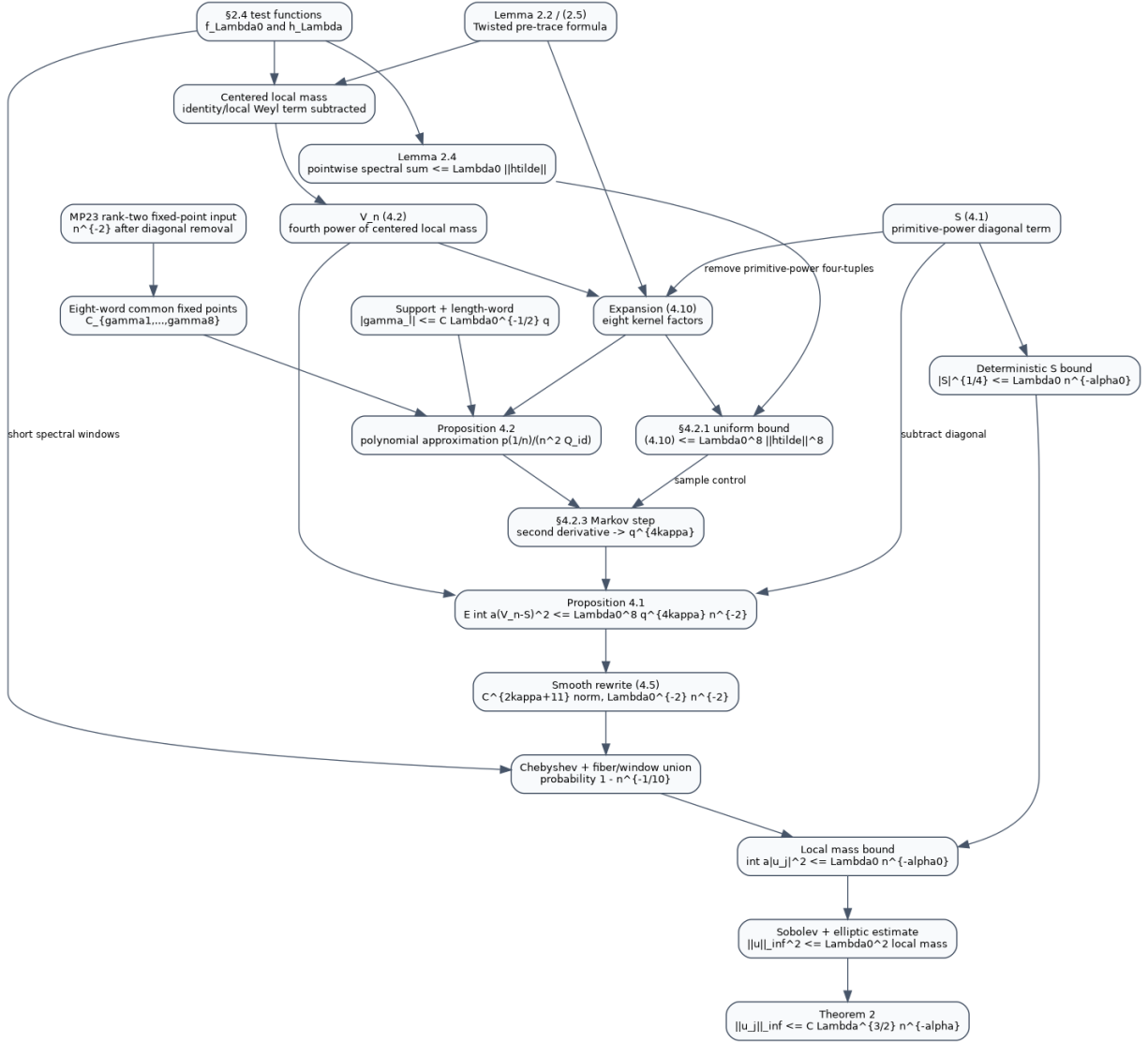


Figure 1: Dependency graph for Theorem 2 delocalization, showing the route from pre-trace formula and fourth-moment control to the final sup-norm estimate.

Cycle 5: M2 Proof-Ledger Closure

The Cycle 5 researcher brief stated that the remaining work was consolidation, not another deep proof audit. The required deliverable was a unified Theorem 1 reconstruction plus a plan-level closure note and cross-proof loss map.

The worker produced five artifacts:

- docs/proof_ledger/rigidity_proof_reconstruction.md
- docs/proof_ledger/m2_loss_map.md
- docs/proof_ledger/m2_proof_ledger_closure.md
- docs/proof_ledger/m2_closure_dependency_graph.dot
- docs/proof_ledger/m2_closure_dependency_graph.png

The closure graph was rendered successfully.

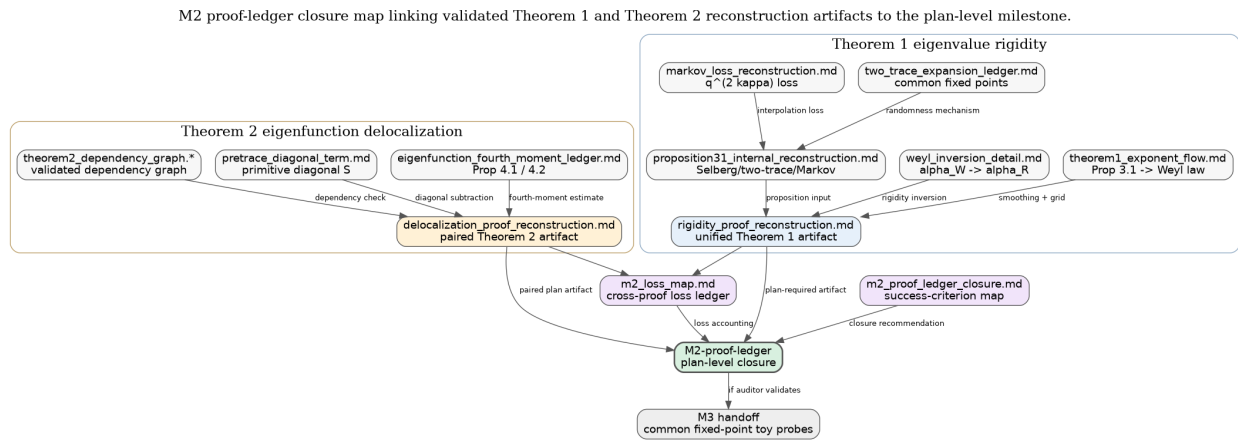


Figure 2: M2 proof-ledger closure map linking validated Theorem 1 and Theorem 2 reconstruction artifacts to the plan-level milestone.

The Cycle 5 audit validated broad M2-proof-ledger, narrowly for local proof reconstruction and quantitative dependency/loss accounting. The audit emphasized that this did not validate M3, M4, M5, or M6, and that imported MPvH, Nau, and MP23 inputs remain black boxes for later external audit.

Cycle 6: First M3 Computational Probe

The Cycle 6 researcher brief moved the campaign to M3-computational-probes. The target was a finite random-permutation common-fixed-point experiment, because the proof ledger had localized the main unresolved mechanism to common-fixed-point statistics and polynomial interpolation rather than direct numerical simulation of hyperbolic spectra.

The worker produced:

- scripts/probe_common_fixed_points.py
- tests/test_permutation_word_eval.py
- data/polynomial_method/common_fixed_point_probe.csv
- data/polynomial_method/common_fixed_point_summary.csv
- reports/figures/m3_common_fixed_point_scaling.png
- reports/figures/m3_common_fixed_point_tails.png
- reports/computational_probes/m3_common_fixed_point_probe.md

The final dataset contains 136,000 rows over 17 word families, four cover degrees, and 2,000 samples per cover degree. The two final figures were verified as readable 1280 x 800 PNGs.

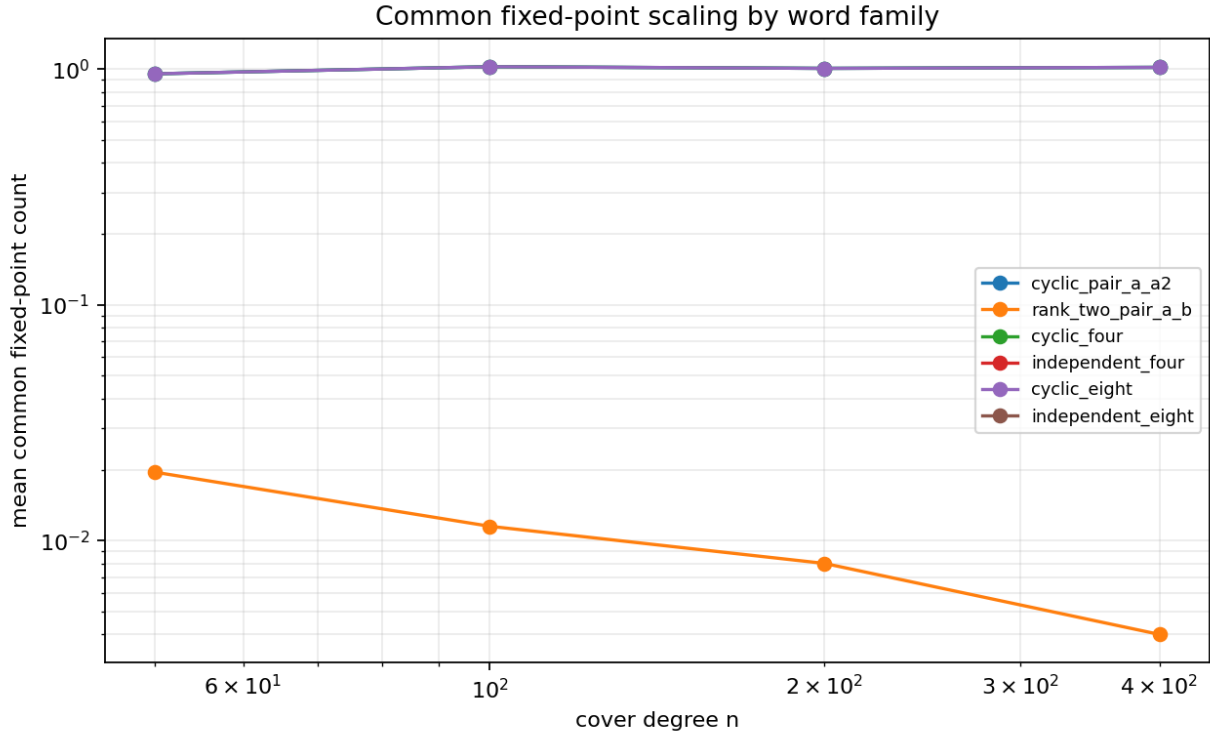


Figure 3: Mean common-fixed-point counts versus cover degree n for cyclic/diagonal and rank-two/noncyclic word families.

The Cycle 6 audit validated the first benchmark and kept M3-computational-probes marked in-progress, not fully validated.

Findings

Finding 1: Theorem 2 Uses the Same Macro-Template as Theorem 1, but With New Local Fourth-Moment Structure

Cycle 4 reconstructed the Theorem 2 pipeline as:

```
twisted pre-trace formula
-> centered local spectral mass
-> fourth power  $V_n$ 
-> subtract primitive-power diagonal  $S$ 
-> Proposition 4.2 eight-word polynomial approximation
-> Markov second-derivative  $q^{\{4 \text{ kappa}\}}$  loss
-> Proposition 4.1
-> Chebyshev, fiber union, and window union
-> local  $L_2$  mass bound
-> Sobolev/elliptic conversion to  $L$ -infinity
```

This parallels Theorem 1 at the highest level: both proofs use compactly supported transforms, trace or pre-trace expansion, polynomial approximation in $1/n$, Markov brothers interpolation, and Chebyshev plus grid union. The difference is that Theorem 2 is local and

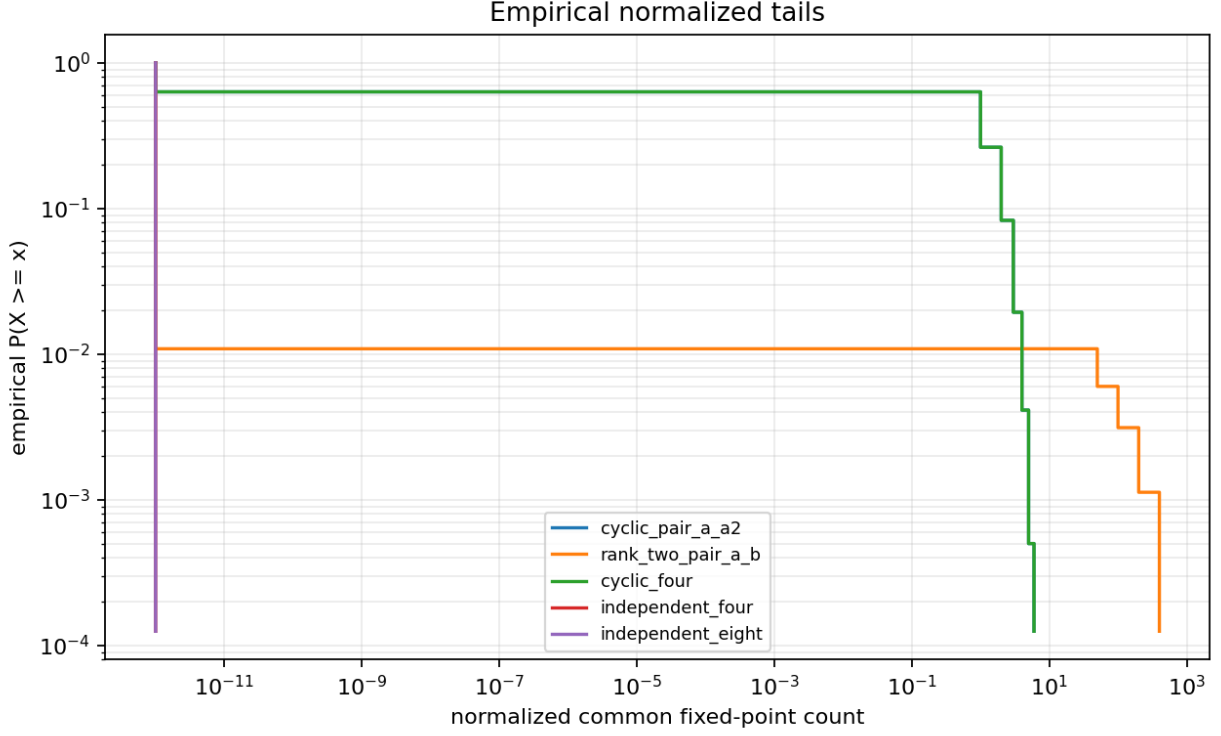


Figure 4: Empirical tail behavior of normalized common-fixed-point counts across toy word families.

fourth-moment based. It introduces S , an eight-word common-fixed-point statistic, a fiber union bound, and a final local-mass-to-pointwise conversion.

Finding 2: The Diagonal Term s Is a Structural Subtraction

The Cycle 4 artifacts define S as the primitive-power diagonal part of the fourth power of the centered pre-trace sum. In local notation, it sums over four nonzero powers of one primitive element. Its purpose is to remove cyclic four-tuples before applying the rank-two common-fixed-point input.

The audit accepted the mechanism: V_n alone contains cyclic primitive-power contributions too large for the rank-two estimate, while $V_n - S$ leaves a non-diagonal statistic to which the eight-word folded-graph and common-fixed-point machinery applies.

Finding 3: The Visible Losses Are Now Separated Across the Two Main Theorems

Cycle 5 produced the cross-proof loss map. The main distinctions are:

Loss source	Location	Role
$q^{2\kappa}$	Proposition 3.1	trace-side Markov interpolation loss
$m = \kappa + 3 + K$	Theorem 1 smoothing	derivative order for smooth cutoff conversion
$\alpha_W \rightarrow \alpha_R$	Weyl inversion	edge loss near $\lambda = 1/4$

Loss source	Location	Role
$q^{4\kappa}$	Proposition 4.1/4.2	pre-trace fourth-moment Markov second-derivative loss
fiber union	Theorem 2 probability conversion	new local-in-fiber union loss
$\Lambda_0^{3/2}$	Theorem 2 final bound	Sobolev/elliptic local-mass-to- L^∞ conversion

The closure audit accepted this accounting and validated M2-proof-ledger.

Finding 4: The First M3 Benchmark Supports the Diagonal-Subtraction Heuristic at Toy Level

Cycle 6 implemented a free-group random-permutation model. For each cover degree n , the script samples random permutations for free generators, evaluates reduced words, and counts

$$\#\{x \in \{1, \dots, n\} : w_1x = \dots = w_kx = x\}.$$

The main numerical means were:

family	n=50	n=100	n=200	n=400
cyclic_pair_a_a2	0.9555	1.0240	1.0075	1.0175
cyclic_four	0.9555	1.0240	1.0075	1.0175
cyclic_eight	0.9555	1.0240	1.0075	1.0175
rank_two_pair_a_b	0.0195	0.0115	0.0080	0.0040
independent_four	0.0000	0.0000	0.0000	0.0000
independent_eight	0.0000	0.0000	0.0000	0.0000

The result supports the proof-ledger heuristic: cyclic primitive-power families retain order-one common fixed-point counts, while the rank-two pair (a, b) is much smaller and scales near $1/n$.

Finding 5: Naive Eight-Word Pointwise Intersections Are Not the Right Next Model

Cycle 6 also produced a useful null result. The rank-two composite families, such as a, b, ab, aB , had the same counts as the pair a, b , because any point fixed by both a and b is automatically fixed by those composites. The audit accepted this as a limitation rather than a failure.

The next M3 probe should therefore move closer to folded-graph/common-quotient statistics or polynomial-window diagnostics, instead of adding more words to a pointwise common fixed set.

Discussion

Cycles 4-6 moved the campaign from proof reconstruction into empirical mechanism testing. The validated proof ledger now covers both main theorems at the local-paper level:

- Theorem 1 is represented by a unified rigidity reconstruction linking Proposition 3.1, smoothing, Chebyshev/grid probability, Weyl law, and rigidity inversion.
- Theorem 2 is represented by a delocalization reconstruction linking the pre-trace formula, V_n , S , Proposition 4.1/4.2, probability conversion, local mass, and the final L^∞ estimate.

The main remaining proof-theoretic caveat is external: MPvH-style embedding expansion, Nau boundedness, and MP23 rank-two common-fixed-point estimates are treated as imported inputs. This was accepted for M2-proof-ledger because that milestone required local proof reconstruction and dependency accounting, not full literature rederivation.

The computational direction is now better defined. M3 should keep the Cycle 6 script as a baseline control suite, but the next substantive observable should encode folded quotient structure rather than only pointwise fixed-set intersections.

Open Questions

1. Can the Markov interpolation losses $q^{2\kappa}$ and $q^{4\kappa}$ be separated into structural and technical components by a sharper polynomial-method analysis?
2. Can a finite folded-graph/common-quotient statistic reproduce the eight-word mechanism behind Proposition 4.2 more faithfully than naive common fixed-point intersections?
3. Can polynomial-window diagnostics be added to the M3 benchmark suite without requiring a full hyperbolic spectral simulation?
4. Are the imported MPvH, Nau, and MP23 inputs sharp enough for the exponents used in Kim-Tao, or are they only convenient black boxes?
5. Can the final Theorem 2 $\Lambda^{3/2}$ behavior be improved by refining the local-mass-to- L^∞ conversion, beyond the interpolation route already noted by the paper?

References

No REFERENCES.md file was present in the workspace during this reporting pass. The report therefore cites only local sources by path and session ID:

- Local paper files: 2603.01127.pdf, 2603.01127.txt.
- Cycle 4 sessions: 9b69e544-cee2-427e-88a7-a2b32be2d4df, 9a8d5232-7da5-416e-be1a-8515bcece60c, dddcaf65-1172-4fa3-a859-9374ae0213b4.
- Cycle 5 sessions: 0bf32ed4-2866-4671-af3c-2f1b6f30f8ba, b5a3c22f-3083-4803-b8c6-1f8e8dee3dc2, 7dc50143-c79a-4b3c-8371-63f8e61aa904.
- Cycle 6 sessions: 40632ca0-1b63-4d2b-9bc8-c8baab1f4ae4, fe7f867c-00e3-47b8-a4cd-172c205ba885, 3229364f-0203-4038-8c82-32992a28250f.

Appendix: Implementation Details

Code and Artifact Organization

Area	Files	Purpose
Theorem 2 proof ledger	docs/proof_ledger/delocalization_proof_reconstruction.md, eigenfunction_fourth_moment_ledger.md, pretrace_diagonal_term.md	Reconstruct the pre-trace/fourth-moment delocalization proof.

Area	Files	Purpose
Theorem 2 graph	docs/proof_ledger/theorem2_dependency_graph.dot, .png	Visual dependency map for Theorem 2.
M2 closure	docs/proof_ledger/rigidity_proof_reconstruction.md, m2_loss_map.md, m2_proof_ledger_closure.md	Consolidate and close the proof-ledger milestone.
M2 closure graph	docs/proof_ledger/m2_closure_dependency_graph.dot, .png	Visual map from validated slices to broad M2-proof-ledger.
M3 probe script	scripts/probe_common_fixed_points.py	Random-permutation common-fixed-point Monte Carlo CLI.
M3 tests	tests/test_permutation_word_eval.py	Direct-run tests for word evaluation and reproducibility.
M3 data	data/polynomial_method/common_fixed_point_probe.csv, common_fixed_point_summary.csv	Final raw and summarized datasets.
M3 report and figures	reports/computational_probes/m3_common_fixed_point_probe.md, reports/figures/m3_common_fixed_point_scaling.png, reports/figures/m3_common_fixed_point_tails.png	Empirical report and plots.

File Counts

File group	Count or lines
Campaign scripts	2
Campaign script lines	342
Campaign test files	1
Campaign test lines	57
Cycle 6 final raw CSV rows	136,000
Cycle 6 word families	17
Cycle 6 final PNG figures	2
Cycle 4-5 new proof-ledger PNG figures	2

Validation Results

Cycle 4 validation:

- docs/proof_ledger/theorem2_dependency_graph.png readable, 1459 x 1343, RGBA.
- promise_check passed with warnings only.
- org_check passed with warnings only.
- Audit decision: VALIDATED, with broad M2-proof-ledger reopened and kept active.

Cycle 5 validation:

- docs/proof_ledger/m2_closure_dependency_graph.png readable, 2125 x 745, RGB.
- promise_check passed with warnings only, events: 19.
- org_check passed with warnings only.
- Audit decision: VALIDATED, closing broad M2-proof-ledger narrowly.

Cycle 6 validation:

- python3 -m py_compile scripts/probe_common_fixed_points.py tests/test_permutation_word_eval.py: passed.
- python3 tests/test_permutation_word_eval.py: passed.
- Smoke CLI run to /tmp: passed.
- Final CSV integrity check: 136000 rows with required columns.
- Final figures readable: both 1280 x 800, RGBA.
- promise_check passed with warnings only, events: 20.
- org_check passed with warnings only.
- Audit decision: VALIDATED, with M3-computational-probes kept in-progress.

Known validator warnings remain non-blocking:

- Pre-existing noncanonical docs/paper_map/ ledger artifact path.
- Future milestones M4, M5, and M6 have no ledger events yet.
- Old cycle 1-3 report files are orphan artifacts in managed paths.
- Root paper/runtime files are outside the organization allow-set.
- Several requested graph figures are under docs/, which org_check warns about but auditors accepted because the briefs explicitly requested those paths.

Session Cross-Reference Map

Cycle	Researcher role	Worker output	Audit decision
4	Directed Theorem 2 pre-trace/fourth-moment reconstruction.	Built Theorem 2 proof ledgers and dependency graph.	Validated slice; fixed premature broad M2 closure.
5	Directed plan-level M2 consolidation.	Built unified rigidity reconstruction, loss map, closure note, and graph.	Validated broad M2 narrowly.
6	Directed first M3 random-permutation probe.	Built script, tests, data, figures, and computational report.	Validated first benchmark; kept M3 in progress.

Manifest Update

MANIFEST.md was replaced with a current snapshot after this reporting pass. It now lists both scripts, the test file, current file counts, cross-references from proof-ledger artifacts into M3, and the milestone state:

- M1-paper-map: validated.
- M2-proof-ledger: validated narrowly for local proof reconstruction and quantitative dependency/loss accounting.
- M3-computational-probes: in-progress, with a validated first benchmark.

- M4-formal-certification: pending.
- M5-extension-candidates: pending.
- M6-final-synthesis: pending.